

MUSKOKA, ON, Canada

WMO: 715320

Lat: 44.973N

Lon: 79.304W

Elev: 925

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-9.9	-24.0	1.5	-14.8	-17.4	2.2	-8.8	20.8	18.6	18.8	17.2	3.2	320	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	85.6	70.3	82.7	68.6	80.1	67.1	72.9	81.4	71.3	78.9	69.6	76.1	9.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	115.8	76.3	68.8	109.6	74.9	67.2	103.6	73.5	37.2	81.8	35.7	79.1	34.3	76.2	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.7	16.5	14.7	-27.5	90.3	6.3	2.2	-32.1	91.9	-35.8	93.2	-39.4	94.5	-44.0	96.1		
			-27.8	76.2	6.3	1.8	-32.3	77.5	-36.0	78.6	-39.5	79.6	-44.0	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	15.2	16.5	26.3	40.4	53.2	62.1	67.1	65.4	58.2	46.1	34.2	22.3
	DBStd	20.61	13.47	11.72	11.85	9.24	8.34	6.58	5.45	5.43	7.65	8.39	9.44	11.31
	HDD50	4658	1079	937	739	312	60	2	0	0	15	180	476	858
	HDD65	8516	1544	1357	1201	738	378	132	42	62	229	587	923	1323
	CDD50	1881	0	0	3	25	159	366	529	476	262	59	2	0
	CDD65	265	0	0	0	1	13	45	106	73	26	1	0	0
	CDH74	2640	0	0	2	20	185	497	1023	666	235	12	0	0
	CDH80	585	0	0	0	2	40	115	240	145	43	0	0	0
Wind	WSAvg	6.7	7.2	7.2	7.4	7.7	6.7	6.0	5.6	5.4	5.6	6.7	7.2	7.3
Precipitation	PrecAvg	43.4	3.8	2.7	2.9	3.1	3.5	3.5	3.1	3.7	4.4	4.0	4.6	4.5
	PrecMax	62.2	5.5	5.7	5.1	4.9	6.9	8.4	7.0	7.0	7.6	6.5	7.8	8.0
	PrecMin	33.7	1.1	0.6	0.6	1.2	0.8	1.3	0.3	1.0	1.9	1.5	2.5	2.0
	PrecStd	6.0	1.1	1.4	1.0	1.1	1.3	1.8	1.6	1.3	1.6	1.5	1.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	45.0	61.8	77.0	84.7	87.3	89.2	88.6	85.1	75.8	60.9	50.1
		MCWB	44.8	42.2	52.4	59.8	67.6	71.2	71.8	71.1	71.2	64.0	52.9	47.6
	2%	DB	40.7	40.0	51.6	68.5	79.0	83.5	85.5	83.9	80.3	69.4	55.6	43.9
		MCWB	39.2	37.5	44.8	53.8	63.5	68.8	70.5	69.2	68.4	59.7	51.3	42.1
	5%	DB	36.5	35.9	46.4	62.0	74.5	79.9	82.6	80.5	76.1	64.4	51.9	39.7
		MCWB	35.0	33.7	41.1	50.4	59.8	66.2	68.6	68.4	65.7	58.8	47.5	38.3
	10%	DB	33.2	33.1	42.1	56.7	69.6	76.0	79.9	77.5	72.4	59.9	48.0	36.3
		MCWB	32.0	31.2	37.7	47.6	57.4	63.8	66.8	66.1	64.2	55.4	45.3	34.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	42.9	56.0	62.4	70.0	74.1	75.5	74.2	72.9	67.1	55.8	48.6
		MCDB	45.6	43.6	61.0	71.3	80.2	83.4	84.5	81.6	80.3	71.8	58.0	49.8
	2%	WB	39.4	37.8	46.5	57.3	66.7	71.5	73.0	72.2	70.8	62.4	51.9	42.5
		MCDB	40.1	39.5	49.6	65.6	75.8	80.4	81.6	79.9	76.9	67.4	55.2	43.6
	5%	WB	35.2	34.0	41.8	52.7	63.3	68.9	71.1	70.5	68.2	59.1	48.2	38.5
		MCDB	36.3	35.3	45.3	59.3	70.1	75.6	78.7	77.0	73.2	63.4	50.9	39.5
	10%	WB	32.2	31.4	38.0	48.5	60.1	66.6	69.4	68.4	65.6	56.0	45.3	35.0
		MCDB	33.1	33.1	41.8	55.5	66.2	72.9	76.1	74.4	70.9	59.0	48.0	36.0
Mean Daily Temperature Range	5% DB	MDBR	17.9	19.4	19.0	20.3	23.1	21.7	21.8	21.5	21.4	17.2	14.7	14.4
		MCDBR	14.6	18.3	21.6	31.3	30.2	27.3	25.9	25.4	25.6	24.4	21.4	13.6
		MCWBR	13.9	16.4	15.4	18.6	15.2	13.7	12.3	12.5	14.2	15.5	16.8	12.7
	5% WB	MCDBR	14.0	16.8	19.0	25.2	23.4	22.3	20.8	20.1	20.9	21.1	18.7	12.8
		MCWBR	13.7	16.0	15.1	17.6	13.8	13.0	11.3	11.7	13.0	14.9	16.2	12.7
Clear-Sky Solar Irradiance	taub	0.287	0.292	0.313	0.354	0.379	0.404	0.406	0.398	0.370	0.343	0.308	0.282	
	taud	2.373	2.342	2.410	2.380	2.354	2.301	2.317	2.332	2.406	2.471	2.516	2.439	
	Ebn at Noon	263	284	293	288	282	274	272	270	270	262	256	254	
	Edh at Noon	24	30	32	36	38	41	39	37	32	26	21	20	
All-Sky Solar Radiation	RadAvg	426	705	1140	1423	1679	1829	1850	1597	1253	734	432	315	
	RadStd	51	88	118	133	167	144	129	89	120	78	56	37	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page